

Project 2: Technical report

1. Introduction

This project builds on Project 1 by preparing the historical stock price dataset for advanced analysis and future machine learning. The main goal was to improve data quality, manage anomalies, create useful features, transform variables, and save a clean modeling-ready dataset.

2. Methods Used and Rationale

The work was completed in Google Colab using Python, Pandas, NumPy, Matplotlib, and Scikit-learn. The dataset was loaded from the ZIP file used in Project 1. Because the stock price file is large, a sample of the price data was used to make the notebook easier to run in Colab.

Missing values: The dataset was checked for missing values. Time-based forward fill and backward fill were included as an appropriate strategy for stock data because values should be handled within each ticker and date order.

Outliers: The Interquartile Range (IQR) method was used to detect extreme values in close price and trading volume. Capping was selected instead of deletion because financial data naturally contains unusual market events that should not be completely removed.

Feature engineering: New variables were created, including price range, daily return, daily return percentage, 7-day rolling average, 30-day rolling average, and rolling volatility. These features capture movement, trend, and risk.

Scaling and encoding: Min-Max normalization and standardization were applied to numerical variables so model inputs have comparable scales. Sector and exchange were one-hot encoded to convert categorical variables into numerical columns.

Data splitting: The final dataset was split into training, validation, and test sets to prepare it for model training, tuning, and final evaluation.

3. Findings

The initial inspection showed that the datasets were complete and did not contain missing values or duplicate records. Outlier detection identified a large number of extreme values in close price and trading volume. After capping, the dataset retained its original records while reducing the influence of extreme observations.

The engineered features improved the dataset by adding information about daily price movement, short-term and long-term trends, and volatility. The rolling average graphs showed that the 7-day average follows price changes more closely, while the 30-day average provides a smoother trend. Rolling volatility helped identify periods of greater market uncertainty.

4. Before-and-After Snapshots

The following snapshots demonstrate the effects of the cleaning and transformation process.

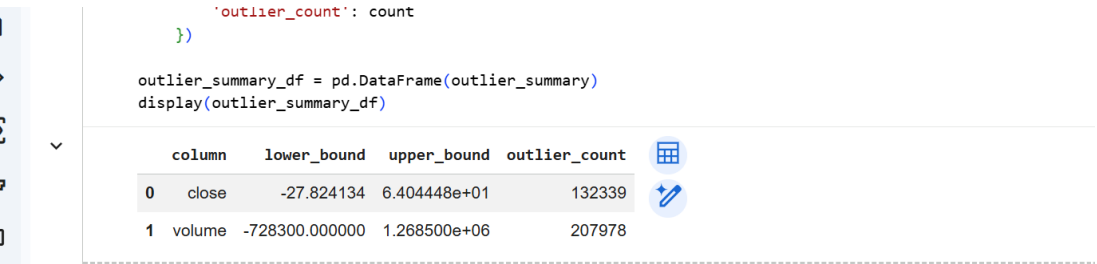


Figure 1. IQR outlier summary for close price and volume.

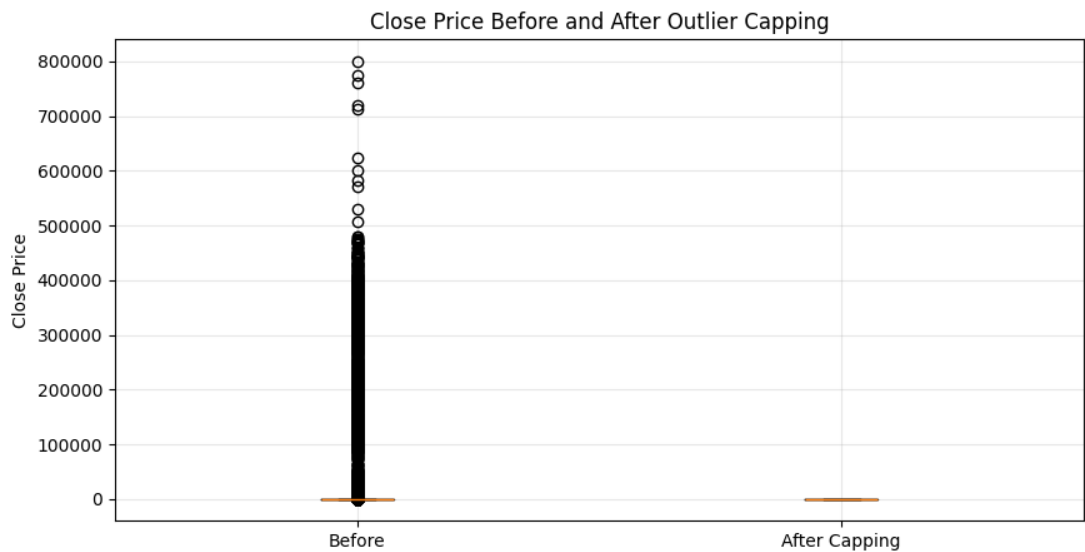


Figure 2. Close price before and after outlier capping.

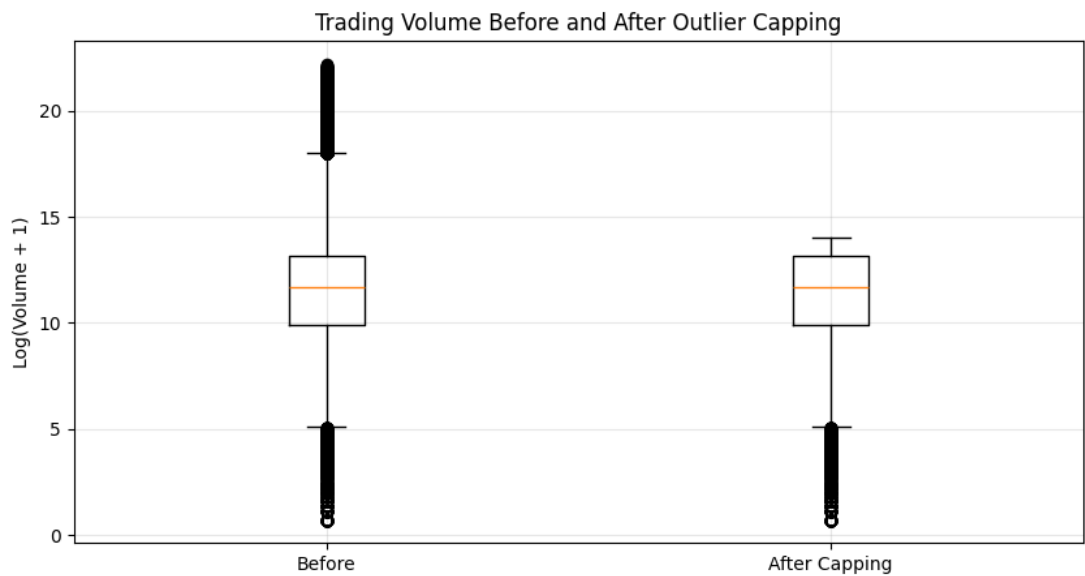


Figure 3. Trading volume before and after outlier capping.

	ticker	date	open	high	low	close	price_range	daily_return	daily_return_pct	rolling_avg_7	rolling_avg_30	volatility
0	AAPL	1980-12-12	0.513393	0.515625	0.513393	0.513393	0.002232	0.000000	0.000000	0.513393	0.513393	0.000
1	AAPL	1980-12-15	0.488839	0.488839	0.486607	0.486607	0.002232	-0.002232	-0.004566	0.500000	0.500000	0.003
2	AAPL	1980-12-16	0.453125	0.453125	0.450893	0.450893	0.002232	-0.002232	-0.004926	0.483631	0.483631	0.002
3	AAPL	1980-12-17	0.462054	0.464286	0.462054	0.462054	0.002232	0.000000	0.000000	0.478237	0.478237	0.002
4	AAPL	1980-12-18	0.475446	0.477679	0.475446	0.475446	0.002232	0.000000	0.000000	0.477679	0.477679	0.002

Figure 4. Final transformed dataset showing cleaned and engineered features.

5. How the Steps Prepare the Dataset for Modeling

The cleaning and transformation steps make the dataset more reliable for advanced modeling. Missing value handling prevents incomplete records from interrupting analysis. Outlier capping reduces the effect of extreme values while still preserving important market behavior. Feature engineering adds variables that may improve predictive power. Scaling ensures that numerical features are comparable, and one-hot encoding makes categorical variables usable by machine learning algorithms. Finally, splitting the data into training, validation, and test sets prepares the dataset for proper model development and evaluation.

6. Conclusion

Project 2 successfully prepared the historical stock price dataset for advanced analysis and machine learning. The dataset was inspected, cleaned, transformed, enhanced with new features, encoded, scaled, split into modeling subsets, and saved for future use. These steps create a stronger foundation for future predictive modeling tasks such as stock price prediction, volatility forecasting, and sector-level financial analysis.